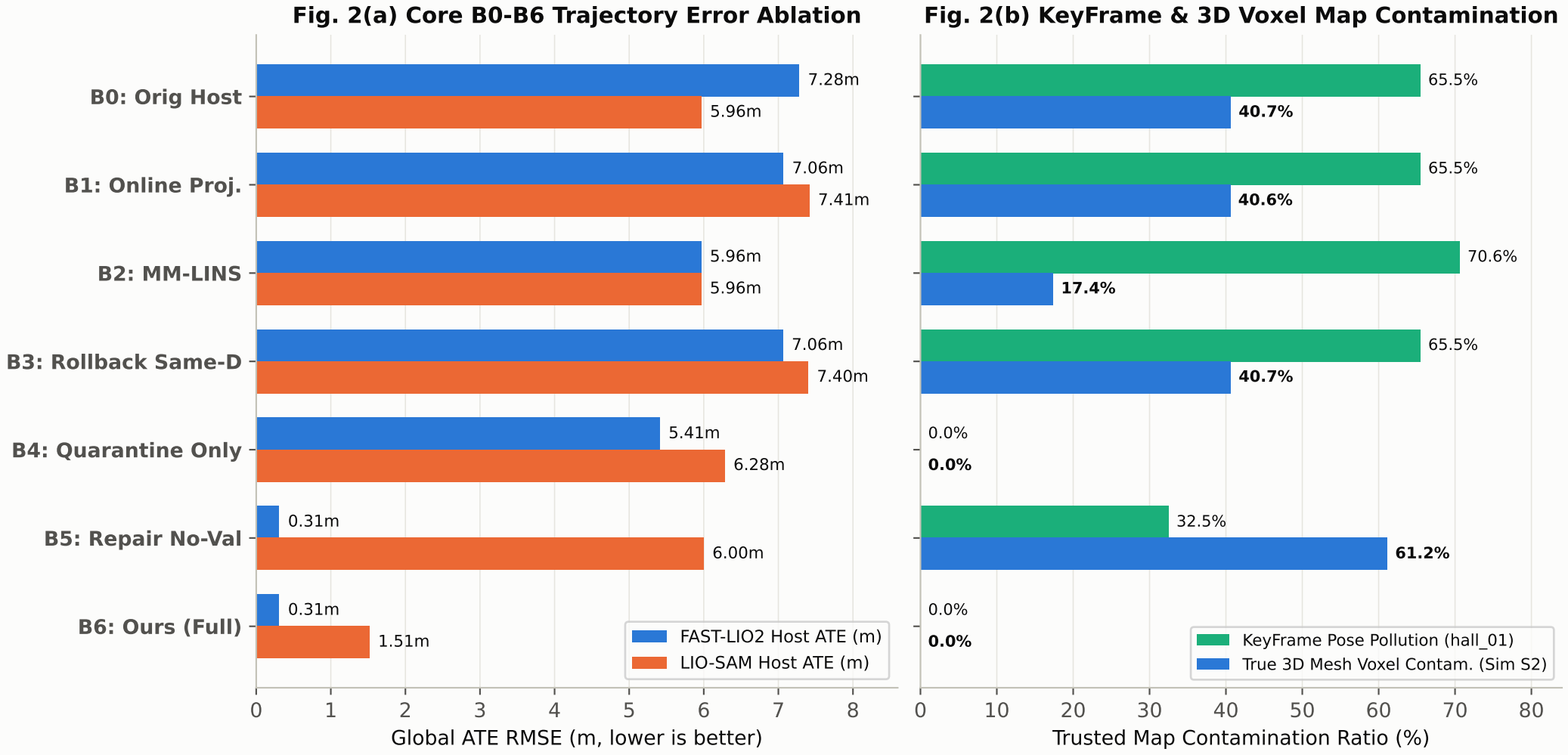
Fig. 2(a) Core B0-B6 Trajectory Error Ablation
B0: Orig Host
B1: Online Proj.
B2: MM-LINS
B3: Rollback Same-D
B4: Quarantine Only
B5: Repair No-Val
B6: Ours (Full)
7.28m
5.96m
7.06m
7.41m
5.96m
5.96m
7.06m
7.40m
5.41m
6.28m
0.31m
6.00m
0.31m
1.51m
FAST-LIO2 Host ATE (m)
LIO-SAM Host ATE (m)
0 1 2 3 4 5 6 7 8
Global ATE RMSE (m, lower is better)
Fig. 2(b) KeyFrame & 3D Voxel Map Contamination
65.5%
40.7%
65.5%
40.6%
70.6%
17.4%
65.5%
40.7%
0.0%
0.0%
32.5%
61.2%
0.0%
0.0%
KeyFrame Pose Pollution (hall_01)
True 3D Mesh Voxel Contam. (Sim S2)
0 10 20 30 40 50 60 70 80
Trusted Map Contamination Ratio (%)